



PlayStations power Air Force's green supercomputer

Condor cluster delivers energy-efficient high-performance computing on the cheap

- By [Henry Kenyon](#)
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The Air Force has long taken an interest in using video games for simulation and modeling, but it's now using their underlying technology for supercomputing.

The Air Force Research Laboratory's Condor Cluster project is using video game console components to make a supercomputer. Built from off-the-shelf components, the guts of the Condor Cluster consist of 1,716 Sony PlayStation 3 game consoles.

Speaking to reporters at a DOD Live Bloggers Roundtable, Mark Barnell, director of high performance computing and the Condor Cluster project at the AFRL, the computer is designed to operate at speeds around half a petaflop, or some 500 trillion floating-point calculations per second. He added that the cluster is currently the 35th or 36th fastest computer in the world. With some tweaks, it could be bumped up to around the 20th fastest machine, he added.

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But raw speed is not one of the Condor Cluster's goals. It's a green machine, designed to demonstrate new ways to use supercomputing resources while using less energy. It is currently the greenest computer in the world, Barnell said.

The computer is also designed to be affordable. The cluster cost \$2 million to build and is much less expensive than general purpose supercomputers, whose prices begin at \$50 million. (The PlayStation 3 sells for \$299 on Amazon. com, so the retail cost of 1,716 of them would be \$513,084.)

Condor's main task is neuromorphic computing — running "learning" algorithms that can teach a machine to read letters, words, symbols and sentences. With such software, a computer could theoretically be instructed to read text and fill in any gaps on its own. The American Forces Press Service [reports](#) that one of the driving goals behind this effort is to have a computer that can take in millions of lines of data and be able to fill in gaps or rearrange material in case of human error.

The cluster can absorb information at about 20 pages per second. Barnell said that even with 20 percent to 30 percent of the characters removed from each page, the computer can recover all of the data with 99.9 percent accuracy. Such a capability would be very useful to Air Force intelligence analysts trying to parse information out of fragmentary documentation.

This is not the first time Sony PlayStations have been used for science. GCN [reported](#) on how scientists at the University of

Massachusetts Dartmouth used a cluster built from eight consoles to model the ripples in space time produced by black holes.

Physics professor Guarav Khanna told GCN that he was attracted to PlayStation 3 because it is an open computing platform and possesses a great deal of raw computing power. Each PS3 processor is equal to about 25 processors in a traditional supercomputer.

About the Author

Henry Kenyon is a contributing writer for Defense Systems.

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